

UNISOLE AI CAMPUS PROGRAM

GROUP 2 — SCIENCE & MATHEMATICS

Pathway 1: Scientific Machine Learning & AI for Science

Target students: Physics • Chemistry • Biology • Applied Science • Other science disciplines

Overview

This pathway is aimed at students who want to apply AI to real scientific problems — simulation, prediction, and experimental data analysis — rather than generic business use cases. It builds from mathematical foundations up through physics-aware AI models, ending in a research-style capstone similar to what a science graduate would produce for a thesis or research internship.

Why this pathway matters: Most AI training treats every dataset the same way. Real science doesn't work that way — physical laws, measurement error, and experimental design all matter. This pathway teaches AI the way a scientist actually needs to use it.

Module 1 — Mathematical Foundations for AI

Every AI model is, underneath, a mathematical object. This module makes that explicit, translating linear algebra and calculus concepts students already know into the language used to describe and optimize models.

Topics covered:

- Linear algebra — Vectors and matrices, Matrix operations, Eigenvalues/eigenvectors
- Calculus fundamentals — Derivatives, Gradients
- Optimization — how models are actually adjusted to fit data
- Probability and Statistics — reasoning under uncertainty

Skills gained: Fluency connecting familiar mathematical concepts directly to how AI models are built and trained.

Module 2 — Scientific Computing

Before modelling, students need fluency with the computational tools scientists actually use daily to process data and solve equations numerically.

Topics covered:

- Python for scientific computing — NumPy, SciPy, Pandas
- Matplotlib — visualizing scientific data
- Jupyter — the standard notebook environment for scientific work
- Numerical methods and Numerical integration — solving equations that can't be solved by hand
- Differential equations — modelling systems that change over time
- Data visualization — communicating results clearly

Skills gained: Practical, hands-on computational fluency that mirrors real research and lab environments.

Module 3 — Machine Learning for Scientists

The core ML toolkit, taught specifically through the lens of experimental and scientific data rather than generic business datasets.

Topics covered:

- ML workflow — Regression, Classification, Clustering
- Feature engineering — turning raw measurements into usable model inputs
- Model evaluation — knowing when a model's predictions can be trusted
- Scientific datasets and Experimental data analysis — working with real, often noisy scientific data

Skills gained: The ability to apply standard ML techniques correctly to experimental data, accounting for its particular challenges (noise, small sample sizes, measurement error).

Module 4 — Deep Learning

Neural network architectures, with attention to how they apply to scientific data types like images, spectra, and time series.

Topics covered:

- Neural networks and Optimization
- CNNs — for image and spatial scientific data
- Sequence modelling — for time-series and signal data
- Transfer learning — adapting existing models to new scientific domains
- Scientific datasets and Model interpretation — understanding *why* a model made a prediction, which matters more in science than in most business contexts

Skills gained: The ability to apply deep learning to scientific data types and interpret results with appropriate scientific rigor.

Module 5 — Scientific Machine Learning

This module introduces the emerging field of Scientific ML — combining traditional physics-based equations with data-driven learning, rather than treating them as separate approaches.

Topics covered:

- Scientific ML concepts and Data-driven modelling
- Physics-based modelling and Hybrid modelling — combining known physics with learned patterns
- Differential equations + ML — solving and learning from equations describing physical systems
- Neural operators — a modern technique for learning solution operators of differential equations
- Scientific prediction — generating predictions grounded in both data and known theory

Skills gained: Exposure to a genuinely cutting-edge, research-active area of AI that most standard courses don't cover at all.

Module 6 — Physics-Informed AI

A specialized and increasingly important technique: building neural networks that are constrained to obey known physical laws, so predictions stay physically plausible even with limited data.

Topics covered:

- Physical constraints and Differential equations
- Loss functions — how physical laws are built directly into model training
- Boundary conditions and Initial conditions — the physical starting points for a simulation
- Physics-Informed Neural Networks (PINNs) — the core technique of this module
- PINN workflow and Scientific model validation

Skills gained: A specialized, high-value skill: building AI models that respect real physical constraints, which is essential in fields like engineering, climate science, and materials research.

Module 7 — Research Computing

The final skillset needed to turn technical work into a credible research output — the same practices used in academic and R&D; settings.

Topics covered:

- Research problem formulation and Literature review
- Dataset creation and Experimental design
- Reproducible computing — making sure results can be verified and repeated
- Git/GitHub for research code
- Scientific documentation, Visualization, Research presentation

Skills gained: The ability to package technical work the way a real research group, journal, or R&D; employer would expect.

Capstone Project

Students choose from capstone directions including: Scientific prediction, Physical-system simulation, Climate/environmental modelling, Computer vision for scientific applications, Physics-informed prediction, Biological/chemical data modelling.

Each capstone follows the full pipeline from problem formulation through to a validated, documented, and presented result — mirroring a genuine research project or R&D; internship deliverable.

Tools & technologies used across this pathway: Python, NumPy, SciPy, Pandas, Matplotlib, Jupyter, neural network frameworks, Physics-Informed Neural Network techniques, Git/GitHub.

Career relevance: Research Assistant, Scientific/Computational Analyst, AI-for-Science roles, R&D; roles in climate, materials, pharma, and engineering-adjacent industries, and strong preparation for postgraduate research.

Student outputs: Research-style project + computational notebook + GitHub/repository + technical report + presentation

Industry-Readiness Layer (*applies alongside this pathway*)

Professional Digital Profile: Professional email, LinkedIn, Resume, Portfolio, Project documentation, Professional communication.

Evidence: Git, GitHub, README, Repository structure, Version control for computational work; Portfolio, Research documentation, and Project presentation for the research side.

Interview Preparation: Resume walkthrough, Tell-me-about-yourself, Project explanation, Behavioural questions, Technical/domain questions, Mock interview, Feedback.

Industry Interaction: Weekly/periodic expert sessions, Career sessions, Industry case studies, How companies recruit, How to approach professionals, Networking etiquette.

Project Defence: Every student must be able to answer — What problem did you solve? Why did you choose this approach? What did you build? What did you learn? What are the limitations? What would you improve?

Final Capstone Framework

1. **Problem** — Identify a real-world problem
2. **Research** — Understand the domain and existing solutions
3. **Data/Information** — Collect and organize relevant information
4. **Build** — Create the appropriate scientific/technical solution
5. **Validate** — Test the solution and interpret results
6. **Document** — Report, Repository/portfolio, Documentation, Presentation
7. **Defend** — Present the project before an evaluator/mentor

What This Student Should Have at the End

Scientific/Computational Project + Notebook/Code + Research Report + Presentation

Training and assessment are program components. Internship, advanced projects, hackathons, research engagement and other opportunities are performance/availability-based and subject to applicable selection procedures.